

Black-Box Regression Estimator for $a'_x = da_x/d\bar{h}$ (no $c(\bar{h})$, no lookup table)

Setting: contrast with the known-wall baseline

	Known-wall (4state_del_hd_ar1)	Black-box (this document)
a_x predict target	$a_{x_{det}}[k] = a_{nom}/c(\bar{h}_d[k])$	none — self-referencing, no external target
a'_x	$-a_x K_h/R$, from the real $c(\bar{h})$	unknown; <i>estimated</i> below
K_h, K'_h	<code>calc_correction_functions</code>	forbidden anywhere, incl. $t=0$ seeding
$\mathbf{F}_e(4, 4)$	λ_c , reverts to $a_{x_{det}}$	driven by $\hat{a}'[k] \cdot \Delta h[k]$ (§below)
Spatial memory	—	none; purely time-recursive (no $\bar{h}$ -indexed map)

Wall *location* is still known ($\bar{h}[k]$ is measurable from $p_m[k]$ and the known wall plane); only the *function* $c(\bar{h})$ is unknown.

Parameter model (black-box)

Define the slow reference level as the existing LP filter of a_{xm} (already in code, pole a_{det_lp}):

$$a_{level}[k] := a_{m_det}[k] = (1 - a_{det_lp}) a_{m_det}[k-1] + a_{det_lp} a_{xm}[k]$$

First-order local expansion around the reference height $\bar{h}_{level}[k]$ that $a_{level}[k]$ corresponds to — *no assumed shape* for $c(\bar{h})$, just a Taylor truism valid for any smooth c :

$$a_x(\bar{h}[k]) \approx a_{level}[k] + a'[k] \Delta h[k], \quad \Delta h[k] := \bar{h}[k] - \bar{h}_{level}[k]$$

Honest proxy: $\delta x_r \approx -\Delta h$

Recall (`pm_to_axm_derivation.tex`): $\delta x_m[k] = x_d[k-d] - p_m[k]$, $\bar{\delta x}_m[k] = (1 - a_{pd})\bar{\delta x}_m[k-1] + a_{pd} \delta x_m[k]$, $\delta x_r[k] = \delta x_m[k] - \bar{\delta x}_m[k]$.

If x_d varies slowly relative to the LP bandwidth a_{pd} (the LP tracks the command almost exactly, $\bar{\delta x}_m[k] \approx x_d[k-d] - \bar{p}_m[k]$), then $x_d[k-d]$ cancels between δx_m and $\bar{\delta x}_m$:

$$\delta x_r[k] \approx -(p_m[k] - \bar{p}_m[k])$$

Projected onto the wall normal, this is exactly the particle's own local height fluctuation, sign-flipped:

$$\delta x_r[k] \approx -\Delta h[k]$$

No $c(\bar{h})$, no $a_{x_{det}}$ — δx_r is already computed by the controller to build a_{xm} .

Caveat. This differs from the known-wall usage of δx_m directly (`aram_variance.tex`: $x_{ram} = -\delta x$). Here δx_m itself carries the slow/deterministic trend $\bar{\delta x}_m$ (tied to x_d 's own shape); using the *residual* δx_r instead avoids conflating the slope signal with the commanded trajectory's shape, which matters once x_d is not near-stationary.

Regression estimator

Substitute $\Delta h \approx -\delta x_r$ into the parameter model, with $n_a[k]$ the a_{xm} chi-squared measurement noise (`pm_to_axm_derivation.tex`):

$$a_{xm}[k] \approx a_{level}[k] - a'[k] \delta x_r[k] + n_a[k]$$

Both sides are already mean-zero after subtracting a_{level} (δx_r is zero-mean by construction, $H_{HP}(z=1) = 0$). The ordinary-least-squares slope is:

$$\hat{a}'[k] = - \frac{\text{Cov}(a_{xm}, \delta x_r)[k]}{\text{Var}(\delta x_r)[k]}$$

Unbiasedness (Isserlis / Wick, third moment)

Lemma (Isserlis, odd order). For a zero-mean jointly Gaussian vector, every odd-order joint moment vanishes: $\mathbb{E}[XYZ] = 0$ for any zero-mean jointly Gaussian (X, Y, Z) , in particular $\mathbb{E}[X^2Y] = 0$ for any zero-mean jointly Gaussian (X, Y) .

$a_{level}[k] = a_{m_det}[k]$ and $n_a[k]$ are both **purely quadratic functionals of past** δx_r — every path from δx_r to a_{xm} or its LP filter goes through $\hat{\sigma}_{\delta x_r}^2[k] = (1 - a_{cov})\hat{\sigma}_{\delta x_r}^2[k-1] + a_{cov}\delta x_r^2[k]$, i.e. a weighted sum of $\delta x_r[j]^2$ for $j \leq k$ (including $j = k$, the current sample). Treating δx_r as (to good approximation) a jointly Gaussian process — the same standing assumption already used for $\text{Var}(\hat{\sigma}_{\delta x_r}^2)$ in `pm_to_axm_derivation.tex` — the Lemma gives, for every j :

$$\text{Cov}(\delta x_r[j]^2, \delta x_r[k]) = \mathbb{E}[\delta x_r[j]^2 \delta x_r[k]] = 0$$

Summing over the EWMA weights that build a_{level} and n_a :

$$\boxed{\text{Cov}(a_{level}, \delta x_r)[k] = 0, \quad \text{Cov}(n_a, \delta x_r)[k] = 0 \quad (\text{exactly, under joint Gaussianity})}$$

Therefore

$$\text{Cov}(a_{xm}, \delta x_r)[k] = -a'[k] \text{Var}(\delta x_r)[k] \implies \mathbb{E}[\hat{a}'[k]] = a'[k].$$

This holds **regardless of a_{cov} , the LP bandwidth a_{pd} , or δx_r 's autocorrelation structure**, and regardless of whether $j = k$ (same-sample) or $j < k$ (past-sample) — the “ a_{xm} is built from δx_r^2 , so correlating it with δx_r is circular” worry vanishes *identically*, not just to first order, given Gaussian δx_r .

Caveat. Real δx_r is only approximately Gaussian (thermal force and sensor noise are Gaussian; the frozen-coefficient / small- a' linearization above is itself an approximation). Residual skew would leak a small bias into $\hat{a}'$; not quantified here.

Real-time recursive form

One new EWMA state (reuses the existing a_{cov} pole — same effective window as $\hat{\sigma}_{\delta x_r}^2$):

$$\boxed{\widehat{\text{Cov}}[k] = (1 - a_{cov}) \widehat{\text{Cov}}[k-1] + a_{cov} (a_{xm}[k] - a_{m_det}[k]) \delta x_r[k]}$$

$$\hat{a}'[k] = -\frac{\widehat{\text{Cov}}[k]}{\hat{\sigma}_{\delta x_r}^2[k]}$$

$\hat{\sigma}_{\delta x_r}^2$ is already computed for a_{xm} — same denominator, no new variance state. Everything on the right-hand side (`a_xm`, `a_m_det`, `dx_r`, `sigma2_dxr_hat`) already exists in `motion_control_law_eq17_4state.m`.

Precision — what does not disappear

Unbiasedness removes the “which $\sigma_{a''}^2$ do I pick” arbitrariness, *not* the SNR problem. From `aram_variance.tex`, single-sample a_{ram} is buried $\sim 73\times$ (std) under a_{xm} ’s own noise, so $\hat{a}'[k]$ needs $N_{\text{eff}} = 1/a_{cov} \gtrsim 73^2 \approx 5300$ effective samples to reach $O(1)$ relative precision — the same dwell-time floor already quantified by the CRLB table in `5state_aprime_selfmod.tex` (near-wall $0.5\text{ s} \rightarrow 81\%$ rel-std, $1.3\text{ s} \rightarrow 50\%$, ...). The exact $\text{Var}(\hat{a}')$ (a further Isserlis expansion, mirroring $\text{Var}(\hat{\sigma}_{\delta x_r}^2)$ ’s derivation) is left as follow-on work; the verification below checks the scaling empirically first.

Verification plan (simple, offline, no controller change)

`diag.a_xm` and `diag.dx_r` are already logged by `motion_control_law_eq17_4state.m`. Both stages are pure post-processing of existing logs — no controller edit needed for a first pass.

V0 (null / bias check). Free-space run ($a' \equiv 0$ everywhere, wall disabled). Compute $\hat{a}'[k]$ via the recursion above from the logged $a_{xm}, \delta x_r$. Expect $\hat{a}'[k] \rightarrow 0$ within sampling noise — confirms the Isserlis cancellation holds in the actual simulated signal, not just the idealized-Gaussian argument.

V1 (oracle recovery). Reuse the existing near-wall 1 Hz oscillation scenario from `aram_variance.tex` (ground-truth $a'(\bar{h}) = -a_x K_h / R$ available from the simulator for comparison only — never fed to the estimator). Compute $\hat{a}'[k]$ the same way; compare to the true $a'(\bar{h}[k])$ as a function of dwell time / effective window ($1/a_{cov}$).

Pass criterion. V0: mean $|\hat{a}'|$ consistent with zero at the sampling-noise level. V1: correlation with the true $a'(\bar{h})$ improves monotonically as a_{cov} is decreased (longer effective window), and the convergence rate matches the order of magnitude of the existing CRLB dwell-time curve.